AI/ML Use Cases in RAN

TELECOM DATA TYPES

- **Data is the fuel for AI/ML**
- **Different data types require different storage methods, processing techniques, ML algorithms, and analysis approaches.**
- **Understanding data is the first step to building successful AI/ML solutions.**

OVERVIEW OF TELECOM DATA TYPES

Telecom Data

Network Data
- PM Data
- Counters
- Traces
- Config (CM)

Customer Data
- CDRs
- Usage
- Profile
- Billing

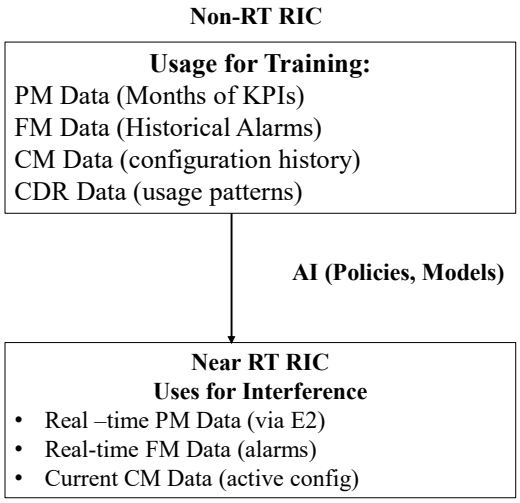
Operational Data
- Alarms (FM)
- Logs
- Tickets
- Inventory

DATA TYPE COMPARISON SUMMARY

Data Type	Format	Volume	Frequency	Primary ML Use
PM Data	Numerical	Very High	15 min	Prediction, Anomaly Detection
FM Data	Events	Variable	Real Time	Root Cause, Fault Prediction
CM Data	Parameters	Low	On Change	Optimisation, Auto Configuration
CDR Data	Records	Extremely High	Per Call/Session	Churn, Fraud, Segmentation
Trace Data	Protocol Messages	Extremely High	Real-Time	Drop Analysis, Protocol Issues
Customer Data	Mixed	Medium	Periodic	Churn, Segment, Personalization

PM : Performance Management
FM : Fault Management
CM : Configuration Management
CDR Data : Call Detail Records
Trace Data : Signalling Traces
Customer Data : Customer Information

DATA TYPES IN O-RAN CONTEXT



Key Takeaways

1. **PM Data is primary source for Network AI/ML**
   - Traffic Prediction, anomaly detection, QoE
2. **FM Data helps with fault management and prediction**
   - Root Cause Analysis, predictive maintenance
3. **CM Data enables network optimization**
   - Parameter Optimization, auto-configuration
4. **CDR Data is good for Customer Insights**
   - Churn Prediction, segmentation, fraud detection
5. **Trace Data provides deep-protocol level insights**
   - Call drops analysis, latency analysis
6. **Customer DATA derives business decisions**
   - Personalization, lifetime value, targeting

Traffic Prediction and Management

- Traffic forecasting using LSTM and time-series models
- Spatial-temporal traffic patterns
- Proactive resource allocation
- Predictive network scaling
- Capacity planning

Mobility and Handover Optimization

- Handover prediction using ML
- Trajectory prediction for connected vehicles
- Cell selection optimization
- Mobility load balancing
- Beam handover in mmWave networks

Network Slicing with AI/ML

- Automated slice creation and configuration
- Slice SLA monitoring and prediction
- Dynamic resource allocation per slice
- URLLC, eMBB, mMTC slice optimization
- Proactive slice scaling

Coverage and Capacity Optimization

- Coverage hole detection
- Traffic hotspot identification
- Cell sleeping for energy efficiency
- Automated neighbor relation (ANR)
- Load balancing using ML

Anomaly Detection and Self-Healing

- Real-time anomaly detection techniques
- Cell outage detection
- Root cause analysis using ML
- Predictive maintenance
- Automated fault recovery

Energy Efficiency Optimization

- AI-driven cell sleeping strategies
- Traffic-aware power control
- Symbol shutdown optimization
- Green AI practices

Physical Layer AI/ML Applications

PHY Layer Intelligence

- Channel estimation using deep learning
- CSI feedback compression with autoencoders
- Beam management and prediction
- MIMO precoding optimization
- MCS (Modulation and Coding Scheme) selection
- Link adaptation using ML

MAC Layer Intelligence

- Intelligent scheduling algorithms
- Resource block allocation using Reinforcement Learning
- Dynamic spectrum sharing
- HARQ optimization
- Power control optimization

Reinforcement Learning in RAN

- RL fundamentals: Q-learning, Policy Gradient
- Dynamic resource allocation using RL
- Multi-agent RL in O-RAN
- Use cases: Spectrum management, load balancing

Federated Learning in O-RAN

- Distributed model training across RAN nodes
- Privacy-preserving collaborative learning
- Benefits and challenges

Generative AI in Telecom

- Large Language Models for network automation
- Intent-based networking
- Synthetic data generation for training
- Code generation for network configuration

6G and AI-Native Networks

- **Digital twins powered by AI**
- **Zero-touch automation**
- **Cognitive networks**
- **AI-as-a-Service in telecom**

Role of ML and DL in 5G (in brief)

1.**Network Optimization:** ML/DL algorithms help in optimizing network parameters like power control, beamforming, and load balancing to enhance performance in 5G and beyond systems.
2.**Resource Allocation:** ML models predict user demand and dynamically allocate resources(e.g.: bandwidth, spectrum) in real-time to improve efficiency and reduce latency.
3.Channel Estimation: DL techniques are used to enhance channel estimation accuracy in complex wireless environments, enabling more reliable communications in 5G/6G systems.
4.**Interference Management:** ML-based approaches are deployed to predict and mitigate interference in highly dense networks, leading to improved network performance.
5.**NetworkSlicing:** ML and DL enable efficient and dynamic network slicing, allowing multiple virtual networks to coexist on a shared physical infrastructure tailored to specific use cases (e.g.: IoT, URLLC).
6.**Self-Organizing Networks (SON):**ML is essential in enabling SONs, where networks can automatically configure, optimize and heal themselves, reducing human intervention.
7.**Energy Efficiency:** ML algorithms optimize energy consumption across network devices, enabling greener and more sustainable 5G and beyond systems.

Role of ML and DL in 5G (in brief)

8.Predictive Maintenance: ML-based predictive analytics help identify and address network faults before they occur, minimizing down time and improving service reliability.
9.Mobility Management: DL models predict user mobility patterns, improving hand over mechanisms and reducing dropped connections in 5G/6G networks.
10.Massive MIMO Optimization: ML/DL algorithms assist in optimizing the configuration and operation of massive MIMO systems for enhanced capacity and spectral efficiency.
11.Beamforming: DL issued for adaptive beamforming, where antenna beams are dynamically adjusted to ensure optimal signals strength and coverage.
12.Security Enhancements: ML/DL techniques help in anomaly detection, intrusion detection and enhancing overall network security in 5G and beyond systems.
13.Traffic Prediction: ML models predict traffic patterns, enabling proactive management of network congestion and ensuring Quality of Service(QoS).
14.Ultra-LowLatency Communication: ML techniques reduce decision-making delays, enabling ultra-low-latency communication, critical for applications like autonomous driving and industrial automation.

Role of ML and DL in 5G (in brief)

15.Edge Computing Integration: ML/DL algorithms deployed at the edge(e.g.: Mobile Edge Computing) enhanced at a processing speed and reduce latency by making decisions closer to the data source, improving real-time application performance in 5G and beyond.
16.Massive IoT Device Management: ML/DL helps manage the massive scale of IoT devices connected to 5G/6G networks by predicting traffic patterns and optimizing connectivity to ensure seamless operation.
17.Cognitive Radio Networks: ML algorithms enable dynamic spectrum access in cognitive radio networks, allowing efficient spectrum utilization by learning and adapting to the surrounding radio environment.
18.Quality of Experience (QoE) Enhancement: ML-driven solutions monitor and improve the end-user experience by predicting and adjusting network parameters based on user behaviour and network conditions.
19.Network Function Virtualization (NFV) Optimization: ML issued to optimize virtualized network functions, making them more efficient by predicting resource needs and balancing loads dynamically

Role of ML and DL in 5G (in brief)

20.Vehicular Communications(V2X): ML/DL models are crucial for enhancing V2X (Vehicle-to-Everything) communications by predicting road conditions, traffic patterns, and ensuring low-latency, reliable communication between vehicles and infrastructure.

21.Fault Detection & Recovery: ML/DL algorithms detect faults in real-time and enable automated recovery processes, improving network resilience and uptime in highly complex 5G/6G architectures.

22.Personalized Services: ML algorithms analyze user data to offer personalized services and network configurations, adapting to user preferences for applications like AR and VR.

23.Intelligent Handover Decisions: ML issued for intelligent handover decisions between different cells and technologies, improving the user experience by reducing the like lihood of dropped calls or interrupted sessions.

Role of ML and DL in 6G and Beyond Systems (in brief)

1.Enhanced Autonomous Networks: In 6G, networks will become more autonomous, driven by AI and ML at every layer. The level of self-organization, self-healing, and self-optimization will be significantly higher compared to 5G.

2.AI-Native Architecture:6G will have AI/ML deeply integrated into the architecture, enabling more proactive and predictive optimization across network layers and nodes. This goes beyond 5G'spartialML-drivenoptimization.

3.THz Communication Optimization: ML/DL will be essential for managing and optimizing Terahertz(THz) communications, which 6G aims to use for ultra-high-speed data transfer over short distances.

4.Holographic Communication: In 6G, ML/DL will be critical form an aging new services like holographic communication and immersive AR/VR experiences, requiring ultra-low latency, massive data processing, and resource allocation.

5.AI-Driven Semantic Communication: 6G is expected to go beyond traditional data transfer to focus on semantic communication, where ML/DL algorithms will interpret and transmit only the"meaning"of data rather than the full raw data, enhancing efficiency.

Role of ML and DL in 6G and Beyond Systems (in brief)

6.Integrated AI/ML at the Edge: ML and Dl will be pushed even further toward the edge in 6G, allowing for real-time decision-making, processing, and optimization at the point of data generation(e.g.: edge devices, base stations).

7.Extreme Latency and Reliability: 6G aims for sub-milli second latencies and ultra-reliable communication (99.99999% reliability). ML will be key to achieving these goals by predicting and managing network behaviour at unprecedented speeds.

8.ML-Driven Privacy & Security: With data privacy and security becoming even more critical in 6G, ML models will be deployed to ensure data protection and secure communication across vast, distributed systems, utilizing federated learning and decentralized ML techniques.

9.Integration with Non-Terrestrial Networks(NTN):In 6G, ML/DL will optimize communication with satellites, drones, and high-altitude platforms, ensuring seamless integration between terrestrial and non-terrestrial networks.

10.Ambient Intelligence:6G is expected to enable ambient intelligence, where smart environments (cities, homes, factories)continuously interact with users.ML/DL will power these systems by interpreting and responding to human behaviour in real-time.

Role of ML and DL in 6G and Beyond Systems (in brief)

11. Sustainability and Green AI: In6G, ML/DL will be focused not just on improving performance but also on minimizing energy consumption and ensuring sustainability across networks, which will be crucial in meeting global green initiatives.

12.Quantum Communication Optimization: As 6G begins to integrate quantum communication technologies, ML/DL will play a role in optimizing quantum channel estimation, error correction, and resource allocation in this new communication paradigm.

13.AI-Enhanced Privacy Mechanisms: Advanced privacy-preserving techniques like homomorphic encryption and differential privacy will be enhanced with AI to protect user's data in massive and distributed 6G systems.

14.Real-TimePredictiveAnalytics:With massive amounts of real-time data generated in 6G,ML and DL will predict network traffic, user demand, and resource utilization with even greater accuracy and speed than in 5G,allowing for more granular control.

15.Human-Centric AI: 6G will emphasize a more human-centric AI, where ML/DL models are designed to understand and adapt to user behaviour and intent, offering personalized services while prioritizing user experience.

Thank You